

ELITE IGCSE ACADEMY

Student Past Paper Solutions

May-Nov 2020 P1HR Worked Solutions

MayNov 2020 | P1HR

27 questions ■ question image followed by worked solution

PREPARED BY

Dr. Eslam Ahmed

Assistant Lecturer, Cairo University Faculty of Engineering

WhatsApp: +20 112 000 9622 | eliteigcse.com

PAST PAPER QUESTION**Q#: 1****Marks: 3**TOPIC: **Fractions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

1 Show that $3\frac{3}{4} \times \frac{7}{9} = 2\frac{11}{12}$

(Total for Question 1 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 1****Marks: 3**TOPIC: **Fractions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Fractions. The tag is correct.**Method**

$$3\frac{3}{4} = \frac{15}{4}$$

Now multiply:

$$\frac{15}{4} \times \frac{7}{9}$$

Cancel 15 and 9:

$$\frac{5}{4} \times \frac{7}{3} = \frac{35}{12}$$

$$\frac{35}{12} = 2\frac{11}{12}$$

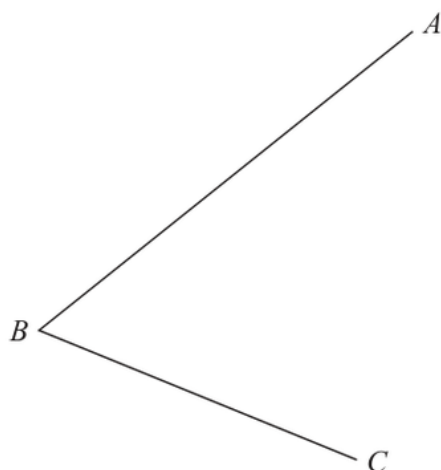
FINAL ANSWER

$$2\frac{11}{12}$$

PAST PAPER QUESTION**Q#: 2****Marks: 2****TOPIC: Bearings, Scale Drawing & Constructions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 2 Using ruler and compasses only, construct the bisector of angle ABC .
You must show all your construction lines.



(Total for Question 2 is 2 marks)

Dr. Eslam

PAST PAPER SOLUTION

Q#: 2

Marks: 2

TOPIC: Bearings, Scale Drawing & Constructions

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Bearings, Scale Drawing & Constructions. The tag is correct.

Construction answer

To construct the bisector of angle ABC :

1. Put the compass point at B and draw an arc cutting BA and BC .
2. From those two cut points, draw equal arcs inside the angle.
3. Draw a straight line from B through the point where the two arcs cross.

FINAL ANSWER

This line bisects angle ABC .

PAST PAPER QUESTION**Q#: 3****Marks: 6****TOPIC: Solving Linear Equations**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

3 (a) Simplify $h^7 \times h^2$
(1)

$$G = c^2 - 4c$$

(b) Find the value of G when $c = -5$

$$G = \dots\dots\dots$$

(2)

(c) Solve $\frac{5x-3}{4} = 2x+3$

Show clear algebraic working.

$$x = \dots\dots\dots$$

(3)

(Total for Question 3 is 6 marks)

Dr. Eslam

PAST PAPER SOLUTION**Q#: 3****Marks: 6****TOPIC: Solving Linear Equations**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to solving linear equations. The original tag was solving quadratic equations, but the equation is linear.

Solution

(a)

$$h^7 \times h^2 = h^{7+2} = h^9$$

(b)

$$G = c^2 - 4c$$

When $c = -5$:

$$G = (-5)^2 - 4(-5) = 25 + 20 = 45$$

(c)

$$\frac{5x - 3}{4} = 2x + 3$$

Multiply by 4:

$$5x - 3 = 8x + 12$$

$$-3x = 15$$

$$x = -5$$

FINAL ANSWER h^9 , $G = 45$, and $x = -5$.

PAST PAPER QUESTION**Q#: 4****Marks: 5**TOPIC: **Statistics Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 4 The table gives information about the length of time, in minutes, that each of 60 students took to travel to school on Monday.

Length of time (t minutes)	Frequency
$0 < t \leq 10$	4
$10 < t \leq 20$	10
$20 < t \leq 30$	15
$30 < t \leq 40$	25
$40 < t \leq 50$	6

- (a) Write down the modal class interval.

.....
(1)

- (b) Work out an estimate for the mean length of time taken by these 60 students to travel to school on Monday.
Give your answer correct to one decimal place.

..... minutes
(4)

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 5**TOPIC: **Statistics Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**

The modal class is the class with the greatest frequency:

$$30 < t \leq 40$$

Use midpoints 5, 15, 25, 35, 45.

$$5(4) + 15(10) + 25(15) + 35(25) + 45(6) = 1690$$

$$\text{estimated mean} = \frac{1690}{60} = 28.166 \dots$$

FINAL ANSWERmodal class $30 < t \leq 40$, estimated mean 28.2 minutes.

PAST PAPER QUESTION**Q#: 5****Marks: 6**TOPIC: **Percentages**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 5 In 2017, the population of a village was 7500
In 2019, the population of the village was 8265

(a) Work out the percentage increase in the population of the village from 2017 to 2019

..... %
(3)

In a sale, normal prices are reduced by 30%
The sale price of a T-shirt was 31.50 euros.

(b) Work out the normal price of the T-shirt.

..... euros
(3)

(Total for Question 5 is 6 marks)

Dr. Eslam

PAST PAPER SOLUTION**Q#: 5****Marks: 6**TOPIC: **Percentages**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to Percentages. Both parts use percentage change.**Method**

The population increase is

$$8265 - 7500 = 765$$

Percentage increase:

$$\frac{765}{7500} \times 100 = 10.2\%$$

The T-shirt sale price is after a 30% reduction, so it is 70% of the normal price.

$$0.70 \times \text{normal price} = 31.50$$

$$\text{normal price} = 31.50 \div 0.70 = 45$$

FINAL ANSWER

(a) 10.2%, (b) 45 euros.

PAST PAPER QUESTION**Q#: 6****Marks: 2**TOPIC: **Coordinate Geometry**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 6 Point A has coordinates $(-3, 11)$
Point B has coordinates $(47, b)$
The midpoint of AB has coordinates $(a, -19)$

Find the value of a and the value of b .

 $a = \dots\dots\dots$ $b = \dots\dots\dots$

(Total for Question 6 is 2 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 6****Marks: 2****TOPIC: Coordinate Geometry**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Coordinate geometry. The tag is correct.**Solution**

$$A = (3, 11), \quad B = (47, b)$$

The midpoint is $(a, 19)$.

For the x-coordinate:

$$a = \frac{3 + 47}{2} = 25$$

For the y-coordinate:

$$19 = \frac{11 + b}{2}$$

$$38 = 11 + b$$

$$b = 27$$

FINAL ANSWER

$$a = 25, b = 27.$$

PAST PAPER QUESTION**Q#: 7****Marks: 4****TOPIC: Standard & Compound Units**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 7 Pedro drove from Toulouse to Montpellier in 2 hours 42 minutes.
He drove at an average speed of 90 km/hour.

Janine drove from Toulouse to Montpellier along the same route as Pedro.
The journey took her 3 hours.

Work out Janine's average speed for the journey.

..... km/hour

(Total for Question 7 is 4 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 7****Marks: 4****TOPIC: Standard & Compound Units**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Standard & Compound Units. The tag is correct.**Solution**

Pedro's time was

$$2 \text{ h } 42 \text{ min} = 2.7 \text{ h}$$

Distance from Toulouse to Montpellier:

$$90 \times 2.7 = 243 \text{ km}$$

Janine took 3 hours, so her average speed was

$$243 \div 3 = 81$$

FINAL ANSWER

81 km/hour.

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Compound Interest & Depreciation**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 8 Harold bought an antique clock for £1200
The clock increased in value by 8% per year.

Find the value of the clock exactly 3 years after Harold bought the clock.
Give your answer correct to the nearest £.

£

(Total for Question 8 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Compound Interest & Depreciation**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to Compound Interest and Depreciation. This is repeated percentage growth for 3 years.

Method

The multiplier for an 8% increase is

$$1.08$$

Value after 3 years:

$$1200(1.08)^3 = 1511.6544$$

Correct to the nearest pound,

$$1511.6544 \approx 1512$$

FINAL ANSWER

1512 pounds

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Standard & Compound Units**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 9 A box is put on a horizontal table.

The face of the box in contact with the table is a square of side 1.5 metres.

The pressure on the table due to the box is 34.8 newtons/m²

Work out the force exerted by the box on the table.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

..... newtons

(Total for Question 9 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 9****Marks: 3****TOPIC: Standard & Compound Units**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Standard & Compound Units. The tag is correct.**Solution**

The area of the square face is

$$1.5 \times 1.5 = 2.25 \text{ m}^2$$

Use

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

So

$$\text{force} = \text{pressure} \times \text{area}$$

$$34.8 \times 2.25 = 78.3$$

FINAL ANSWER

78.3 newtons.

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC: Percentages**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

10 Alex makes 80 cakes to sell.

He makes chocolate cakes, lemon cakes and fruit cakes where

$$\begin{array}{ccccc} \text{number of} & : & \text{number of} & : & \text{number of} \\ \text{chocolate cakes} & & \text{lemon cakes} & & \text{fruit cakes} \end{array} = 3 : 2 : 5$$

Alex sells

all of the chocolate cakes

 $\frac{3}{4}$ of the lemon cakes $\frac{7}{8}$ of the fruit cakes

The profit he makes on each cake he sells is shown in the table.

Type of cake	Profit per cake he sells
chocolate	£2.00
lemon	£1.70
fruit	£2.40

Work out the total profit that Alex makes from the cakes he sells.

£

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 5****TOPIC: Percentages**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Percentages. The tag is correct, with fractions and ratio used inside the profit calculation.

Method

The ratio parts are

$$3 + 2 + 5 = 10$$

Numbers of cakes made:

$$\text{chocolate} = 80 \times \frac{3}{10} = 24$$

$$\text{lemon} = 80 \times \frac{2}{10} = 16$$

$$\text{fruit} = 80 \times \frac{5}{10} = 40$$

Cakes sold:

$$\text{chocolate} = 24, \quad \text{lemon} = \frac{3}{4} \times 16 = 12, \quad \text{fruit} = \frac{7}{8} \times 40 = 35$$

Total profit:

$$24(2.00) + 12(1.70) + 35(2.40) = 48 + 20.40 + 84 = 152.40$$

FINAL ANSWER

152.40 pounds

PAST PAPER QUESTION**Q#: 11****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

11 The frequency table gives information about the ages of the 80 people in a train carriage.

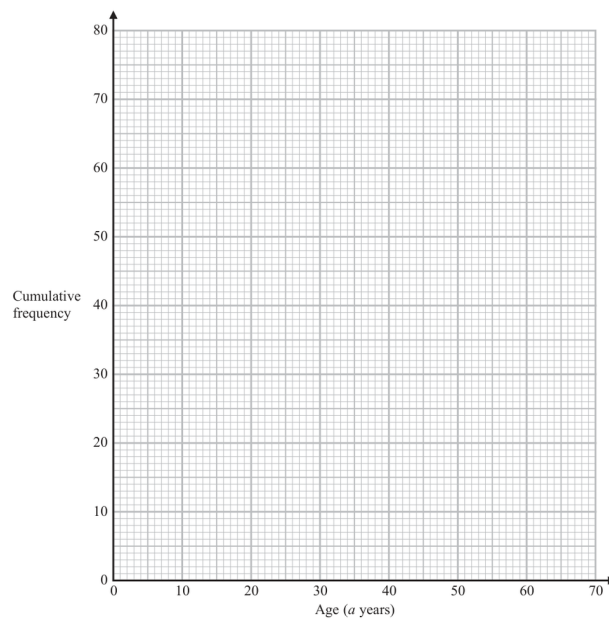
Age (a years)	Frequency
$0 < a \leq 20$	9
$20 < a \leq 30$	19
$30 < a \leq 40$	17
$40 < a \leq 50$	18
$50 < a \leq 60$	13
$60 < a \leq 70$	4

(a) Complete the cumulative frequency table.

Age (a years)	Cumulative frequency
$0 < a \leq 20$	
$0 < a \leq 30$	
$0 < a \leq 40$	
$0 < a \leq 50$	
$0 < a \leq 60$	
$0 < a \leq 70$	

(1)

(b) On the grid, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the median age of the people in the train carriage.

..... years
(2)

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Cumulative Frequency Diagrams. The tag is correct.**Solution**

The cumulative frequencies are

9, 28, 45, 63, 76, 80

The median is the 40th value.

$$30 + \frac{40 - 28}{45 - 28} \times 10 \approx 37.1$$

FINAL ANSWER

median about 37 years.

PAST PAPER QUESTION**Q#: 12****Marks: 4****TOPIC: Simultaneous Equations**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

12 Solve the simultaneous equations

$$7x + 2y = 5.5$$

$$3x - 5y = 17$$

Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$

(Total for Question 12 is 4 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 12****Marks: 4****TOPIC: Simultaneous Equations**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Simultaneous equations. The tag is correct.**Solution**

$$7x + 2y = 5.5$$

$$3x - 5y = 17$$

Multiply the first equation by 5:

$$35x + 10y = 27.5$$

Multiply the second equation by 2:

$$6x - 10y = 34$$

Add:

$$41x = 61.5$$

$$x = 1.5$$

Substitute into $7x + 2y = 5.5$:

$$10.5 + 2y = 5.5$$

$$y = -2.5$$

FINAL ANSWER

$$x = 1.5, y = -2.5.$$

PAST PAPER QUESTION**Q#: 13****Marks: 6****TOPIC: Differentiation**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

13 The curve **C** has equation $y = 5x^3 - x^2 - 6x + 4$ (a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots$$

(2)

There are two points on the curve **C** at which the gradient of the curve is 2(b) Find the x coordinate of each of these two points.
Show clear algebraic working.

\dots\dots\dots

(4)

(Total for Question 13 is 6 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 6****TOPIC: Differentiation**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct. This is finding where the gradient has a given value.**Solution**

$$y = 5x^3 - x^2 - 6x + 4$$

$$\frac{dy}{dx} = 15x^2 - 2x - 6$$

The gradient is 2, so

$$15x^2 - 2x - 6 = 2$$

$$15x^2 - 2x - 8 = 0$$

$$x = \frac{2 \pm \sqrt{(-2)^2 - 4(15)(-8)}}{30}$$

$$x = \frac{2 \pm 22}{30}$$

$$x = \frac{4}{5} \quad \text{or} \quad x = -\frac{2}{3}$$

FINAL ANSWER

$$x = -\frac{2}{3} \text{ and } x = \frac{4}{5}.$$

PAST PAPER QUESTION**Q#: 14****Marks: 3****TOPIC: Expanding Brackets**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

14 Expand and simplify $(4x + 1)(x - 3)(5x + 6)$

(Total for Question 14 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 14****Marks: 3****TOPIC: Expanding Brackets**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Expanding brackets. The tag is correct.**Solution**

$$(4x + 1)(x - 3)(5x + 6)$$

$$(x - 3)(5x + 6) = 5x^2 - 9x - 18$$

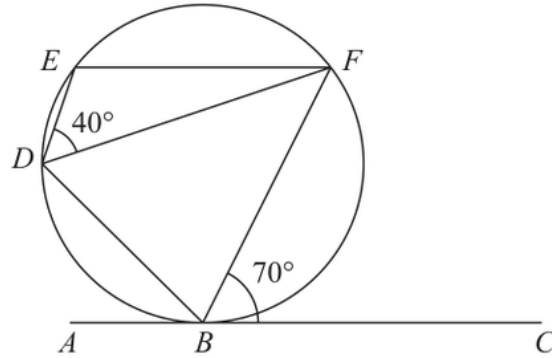
$$(4x + 1)(5x^2 - 9x - 18) = 20x^3 - 31x^2 - 81x - 18$$

FINAL ANSWER

$$20x^3 - 31x^2 - 81x - 18$$

PAST PAPER QUESTION**Q#: 15****Marks: 4****TOPIC: Circle Theorems**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

15Diagram **NOT**
accurately drawn

B, D, E and F are points on a circle.
 ABC is the tangent to the circle at B .

Angle $EDF = 40^\circ$

Angle $FBC = 70^\circ$

Prove that the tangent ABC is parallel to EF .
 Give a reason for each stage of your working.

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION**Q#: 15** **Marks: 4**TOPIC: **Circle Theorems**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Circle Theorems. The tag is correct.**Proof**

By the alternate segment theorem, the angle between tangent BC and chord BF equals the angle in the opposite segment:

$$\angle BEF = 70^\circ$$

Given

$$\angle EDF = 40^\circ$$

Angles EDF and EBF stand on the same chord EF , so

$$\angle EBF = 40^\circ$$

In triangle BEF ,

$$\angle EFB = 180^\circ - 70^\circ - 40^\circ = 70^\circ$$

So

$$\angle EFB = \angle FBC = 70^\circ$$

These are alternate interior angles, so

$$EF \parallel ABC$$

FINAL ANSWER

The tangent ABC is parallel to EF .

PAST PAPER QUESTION**Q#: 16****Marks: 9****TOPIC: Functions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

16 The functions f and g are defined as

$$f: x \mapsto 5x - 7$$

$$g: x \mapsto \frac{5x}{x + 4}$$

(a) Write down the value of x that must be excluded from any domain of g
(1)(b) Find $gf(2.6)$
(2)(c) Solve $fg(x) = 2$ $x =$
(3)(d) Express the inverse function g^{-1} in the form $g^{-1}: x \mapsto \dots$ $g^{-1}: x \mapsto$
(3)**(Total for Question 16 is 9 marks)**

PAST PAPER SOLUTION**Q#: 16****Marks: 9****TOPIC: Functions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Functions. The tag is correct.**Solution**

$$f(x) = 5x - 7, \quad g(x) = \frac{5x}{x+4}$$

(a) The denominator of g cannot be zero, so

$$x = -4$$

(b)

$$f(2.6) = 5(2.6) - 7 = 6$$

$$gf(2.6) = g(6) = \frac{5(6)}{6+4} = 3$$

(c)

$$fg(x) = 2$$

$$f(g(x)) = 2$$

$$5g(x) - 7 = 2$$

$$g(x) = \frac{9}{5}$$

$$\frac{5x}{x+4} = \frac{9}{5}$$

$$25x = 9x + 36$$

$$16x = 36$$

$$x = \frac{9}{4}$$

(d) Let

$$y = \frac{5x}{x+4}$$

$$y(x+4) = 5x$$

$$yx + 4y = 5x$$

$$x(y - 5) = -4y$$

$$x = \frac{4y}{5 - y}$$

So

$$g^{-1}(x) = \frac{4x}{5 - x}$$

FINAL ANSWER

$-4, 3, x = \frac{9}{4}$, and $g^{-1}(x) = \frac{4x}{5 - x}$.

Dr. Eslam Ahmed

PAST PAPER QUESTION

Q#: 17

Marks: 6

TOPIC: 3D Pythagoras & Trigonometry

May-Nov 2020 P1HR - MayNov 2020 | P1HR

17 The diagram shows a prism $ABCDEFGH$ with a horizontal base.

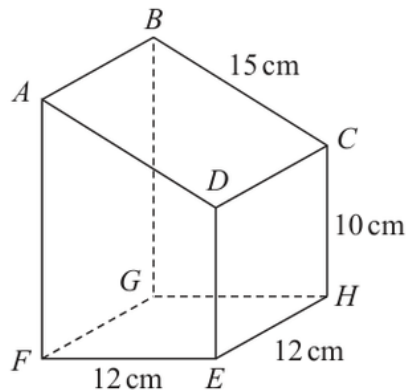


Diagram **NOT**
accurately drawn

The base of the prism, $EFGH$, is a square of side 12 cm.

Trapezium $ADEF$ is a cross section of the prism where AF and DE are vertical edges.

$$DE = CH = 10 \text{ cm}$$

$$AD = BC = 15 \text{ cm}$$

- (a) Work out the size of the angle between CF and the base $EFGH$.
Give your answer correct to one decimal place.

(3)

- (b) Work out the length of BE .
Give your answer correct to one decimal place.

(3)

(Total for Question 17 is 6 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 6****TOPIC: 3D Pythagoras & Trigonometry**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** 3D Pythagoras & Trigonometry. The tag is correct.**Solution**(a) C is vertically above H , so the vertical height from F to C is 10 cm.The projection of CF on the base is FH :

$$FH = 12\sqrt{2}$$

$$\tan \theta = \frac{10}{12\sqrt{2}}$$

$$\theta = 30.508\dots^\circ$$

FINAL ANSWER

30.5°.

(b) In the trapezium cross-section, $AD = 15$ and the horizontal distance is 12, so the difference in vertical heights is

$$\sqrt{15^2 - 12^2} = 9$$

Since $DE = 10$, the height $BG = 19$ cm.The projection of BE on the base is $GE = 12\sqrt{2}$, so

$$BE = \sqrt{19^2 + (12\sqrt{2})^2} = 25.475\dots$$

FINAL ANSWER

25.5 cm.

PAST PAPER QUESTION

Q#: 18

Marks: 4

TOPIC: **Sequences**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 18** In an arithmetic series, the 6th term is 39
In the same arithmetic series, the 19th term is 7.8
Work out the sum of the first 25 terms of the arithmetic series.

(Total for Question 18 is 4 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 18****Marks: 4**TOPIC: **Sequences**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Sequences. The tag is correct.**Solution**Let the first term be a and the common difference be d .

The 6th term is 39:

$$a + 5d = 39$$

The 19th term is 7.8:

$$a + 18d = 7.8$$

Subtract:

$$13d = -31.2$$

$$d = -2.4$$

Then

$$a + 5(-2.4) = 39$$

$$a = 51$$

The sum of the first 25 terms is

$$S_{25} = \frac{25}{2} (2a + 24d)$$

$$S_{25} = \frac{25}{2} (102 - 57.6) = 555$$

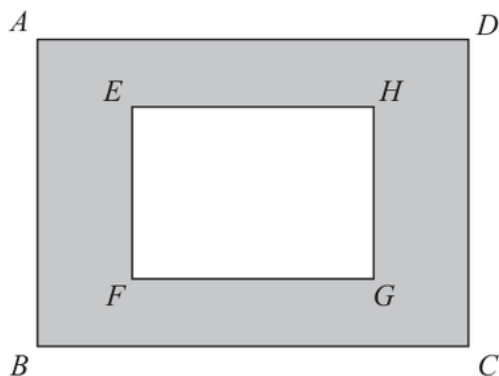
FINAL ANSWER

555.

PAST PAPER QUESTION**Q#: 19** **Marks: 3****TOPIC: Rounding, Estimation & Bounds**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 19** The diagram shows rectangle $ABCD$ with rectangle $EFGH$ cut out to form the shaded region.

Diagram **NOT**
accurately drawn $AD = 8.3$ cm correct to one decimal place $DC = 7.2$ cm correct to one decimal place $EH = 6.2$ cm correct to one decimal place $HG = 5.3$ cm correct to one decimal place

Work out the upper bound of the area of the shaded region.
Show your working clearly.

..... cm^2 **(Total for Question 19 is 3 marks)**

PAST PAPER SOLUTION**Q#: 19****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rounding, Estimation and Bounds. The tag is correct.**Method**

For the upper bound of the shaded area, use the upper bound of the outer rectangle and subtract the lower bound of the cut-out rectangle.

Outer rectangle:

$$AD < 8.35, \quad DC < 7.25$$

Cut-out rectangle:

$$EH \geq 6.15, \quad HG \geq 5.25$$

$$\text{upper shaded area} = 8.35(7.25) - 6.15(5.25)$$

$$= 60.5375 - 32.2875 = 28.25$$

FINAL ANSWER

$$28.25 \text{ cm}^2$$

PAST PAPER QUESTION**Q#: 20****Marks: 2****TOPIC: Transformations of Graphs**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

20 A curve has equation $y = f(x)$

There is only one maximum point on the curve.

The coordinates of this maximum point are $(-3, 4)$

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x) - 6$

(.....,)

(ii) $y = f(2x)$

(.....,)

(Total for Question 20 is 2 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 2****TOPIC: Transformations of Graphs**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**The maximum point of $y = f(x)$ is $(-3, 4)$.

For

$$y = f(x) - 6$$

the graph moves 6 units down:

$$(-3, 4) \rightarrow (-3, -2)$$

For

$$y = f(2x)$$

the x -coordinate is divided by 2:

$$(-3, 4) \rightarrow \left(-\frac{3}{2}, 4\right)$$

FINAL ANSWER $(-3, -2)$ and $\left(-\frac{3}{2}, 4\right)$

PAST PAPER QUESTION**Q#: 20****Marks: 2****TOPIC: Transformations of Graphs**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

20 A curve has equation $y = f(x)$

There is only one maximum point on the curve.

The coordinates of this maximum point are $(-3, 4)$

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x) - 6$

(.....,)

(ii) $y = f(2x)$

(.....,)

(Total for Question 20 is 2 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 2****TOPIC: Transformations of Graphs**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**The maximum point of $y = f(x)$ is $(-3, 4)$.

For

$$y = f(x) - 6$$

the graph moves 6 units down:

$$(-3, 4) \rightarrow (-3, -2)$$

For

$$y = f(2x)$$

the x -coordinate is divided by 2:

$$(-3, 4) \rightarrow \left(-\frac{3}{2}, 4\right)$$

FINAL ANSWER $(-3, -2)$ and $\left(-\frac{3}{2}, 4\right)$

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Algebraic Roots & Indices**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

21 Given that $M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of n for which $M = 2$

(Total for Question 21 is 5 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 21****Marks: 5**TOPIC: **Algebraic Roots & Indices**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Algebraic roots and indices. The tag is correct.**Solution**

$$18^{4n} = (2 \times 3^2)^{4n} = 2^{4n} 3^{8n}$$

$$12^2 = (2^2 \times 3)^2 = 2^4 3^2$$

So

$$M = \frac{2^{4n} 3^{8n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{2^4 3^2}$$

Combine powers of 2:

$$2^{4n+3n^2-18n-4} = 2^{3n^2-14n-4}$$

Combine powers of 3:

$$3^{8n+2-8n-2} = 3^0 = 1$$

So

$$M = 2^{3n^2-14n-4}$$

Given $M = 2$,

$$3n^2 - 14n - 4 = 1$$

$$3n^2 - 14n - 5 = 0$$

$$(3n + 1)(n - 5) = 0$$

FINAL ANSWER

$$n = -\frac{1}{3} \text{ or } n = 5$$

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Algebraic Roots & Indices**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

21 Given that $M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of n for which $M = 2$

(Total for Question 21 is 5 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 21****Marks: 5**TOPIC: **Algebraic Roots & Indices**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$$

Write everything in powers of 2 and 3:

$$18^{4n} = (2 \cdot 3^2)^{4n} = 2^{4n} 3^{8n}$$

$$2^{3(n^2-6n)} = 2^{3n^2-18n}$$

$$3^{2(1-4n)} = 3^{2-8n}$$

$$12^2 = (2^2 \cdot 3)^2 = 2^4 3^2$$

So

$$M = 2^{4n+3n^2-18n-4} \times 3^{8n+2-8n-2}$$

$$M = 2^{3n^2-14n-4}$$

Given $M = 2$,

$$2^{3n^2-14n-4} = 2^1$$

$$3n^2 - 14n - 4 = 1$$

$$3n^2 - 14n - 5 = 0$$

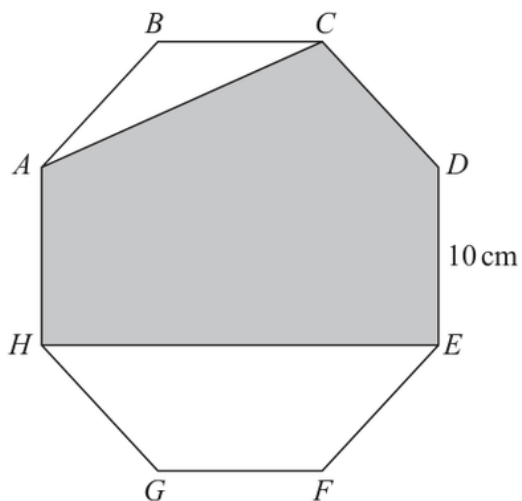
$$(3n + 1)(n - 5) = 0$$

FINAL ANSWER

$$n = 5 \text{ or } n = -\frac{1}{3}$$

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Area & Perimeter**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

22 The diagram shows a regular octagon $ABCDEFGH$.Diagram **NOT**
accurately drawn

Each side of the octagon has length 10 cm.

Find the area of the shaded region $ACDEH$.
Give your answer correct to the nearest cm^2 cm^2 **(Total for Question 22 is 6 marks)**

PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: Area & Perimeter**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

Each side of the regular octagon is 10 cm.

For a regular octagon with side length s ,

$$\text{area} = 2(1 + \sqrt{2})s^2$$

So the area of the whole octagon is

$$2(1 + \sqrt{2})(10)^2 = 200 + 200\sqrt{2}$$

The unshaded triangle ABC has

$$AB = 10, \quad BC = 10, \quad \angle ABC = 135^\circ$$

So

$$\text{area of triangle } ABC = \frac{1}{2}(10)(10) \sin 135^\circ = 25\sqrt{2}$$

At the bottom, the unshaded trapezium $HEFG$ has parallel sides

$$HE = 10 + 10\sqrt{2}, \quad GF = 10$$

and height

$$5\sqrt{2}$$

So

$$\text{area of trapezium } HEFG = \frac{1}{2}(10 + 10 + 10\sqrt{2})(5\sqrt{2})$$

$$= 50 + 50\sqrt{2}$$

The shaded area is

$$(200 + 200\sqrt{2}) - 25\sqrt{2} - (50 + 50\sqrt{2})$$

$$= 150 + 125\sqrt{2}$$

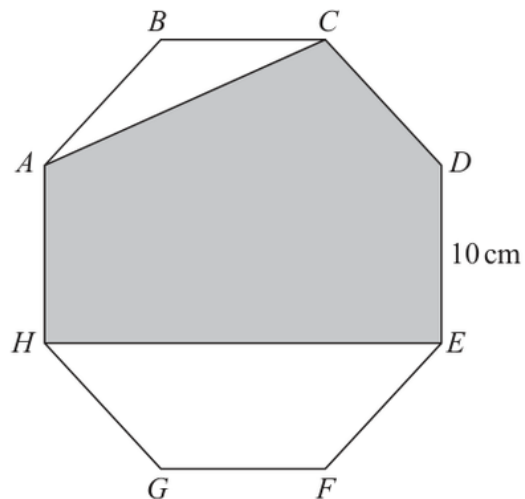
$$= 326.7766 \dots$$

FINAL ANSWER

$$327 \text{ cm}^2$$

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Area & Perimeter**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

22 The diagram shows a regular octagon $ABCDEFGH$.Diagram **NOT**
accurately drawn

Each side of the octagon has length 10 cm.

Find the area of the shaded region $ACDEH$.
Give your answer correct to the nearest cm^2 cm^2 **(Total for Question 22 is 6 marks)**

PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: Area & Perimeter**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

Each side of the regular octagon is 10 cm.

For a regular octagon with side length s ,

$$\text{area} = 2(1 + \sqrt{2})s^2$$

So the area of the whole octagon is

$$2(1 + \sqrt{2})(10)^2 = 200 + 200\sqrt{2}$$

The unshaded triangle ABC has

$$AB = 10, \quad BC = 10, \quad \angle ABC = 135^\circ$$

So

$$\text{area of triangle } ABC = \frac{1}{2}(10)(10) \sin 135^\circ = 25\sqrt{2}$$

At the bottom, the unshaded trapezium $HEFG$ has parallel sides

$$HE = 10 + 10\sqrt{2}, \quad GF = 10$$

and height

$$5\sqrt{2}$$

So

$$\text{area of trapezium } HEFG = \frac{1}{2}(10 + 10 + 10\sqrt{2})(5\sqrt{2})$$

$$= 50 + 50\sqrt{2}$$

The shaded area is

$$(200 + 200\sqrt{2}) - 25\sqrt{2} - (50 + 50\sqrt{2})$$

$$= 150 + 125\sqrt{2}$$

$$= 326.7766 \dots$$

FINAL ANSWER

$$327 \text{ cm}^2$$

PAST PAPER QUESTION

Q#: 23

Marks: 4

TOPIC: **Probability Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

23 In a bag, there are only

3 blue beads
4 white beads
and x orange beads.

Jean is going to take at random two beads from the bag.

The probability that Jean will take two beads of the same colour is $\frac{3}{8}$

Find the total number of beads in the bag.

Show clear algebraic working.

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Probability Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

The total number of beads is

$$x + 7$$

The probability of choosing two beads of the same colour is

$$\frac{\binom{3}{2} + \binom{4}{2} + \binom{x}{2}}{\binom{x+7}{2}}$$

This is equal to $\frac{3}{8}$, so

$$\frac{3 + 6 + \frac{x(x-1)}{2}}{\frac{(x+7)(x+6)}{2}} = \frac{3}{8}$$

$$\frac{x^2 - x + 18}{(x+7)(x+6)} = \frac{3}{8}$$

$$8(x^2 - x + 18) = 3(x+7)(x+6)$$

$$8x^2 - 8x + 144 = 3x^2 + 39x + 126$$

$$5x^2 - 47x + 18 = 0$$

$$(x-9)(5x-2) = 0$$

Since x must be a whole number,

$$x = 9$$

So the total number of beads is

$$9 + 7 = 16$$

FINAL ANSWER

16 beads

PAST PAPER QUESTION

Q#: 23

Marks: 4

TOPIC: **Probability Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

23 In a bag, there are only

3 blue beads
4 white beads
and x orange beads.

Jean is going to take at random two beads from the bag.

The probability that Jean will take two beads of the same colour is $\frac{3}{8}$

Find the total number of beads in the bag.

Show clear algebraic working.

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Probability Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

The total number of beads is

$$x + 7$$

The probability of choosing two beads of the same colour is

$$\frac{\binom{3}{2} + \binom{4}{2} + \binom{x}{2}}{\binom{x+7}{2}}$$

This is equal to $\frac{3}{8}$, so

$$\frac{3 + 6 + \frac{x(x-1)}{2}}{\frac{(x+7)(x+6)}{2}} = \frac{3}{8}$$

$$\frac{x^2 - x + 18}{(x+7)(x+6)} = \frac{3}{8}$$

$$8(x^2 - x + 18) = 3(x+7)(x+6)$$

$$8x^2 - 8x + 144 = 3x^2 + 39x + 126$$

$$5x^2 - 47x + 18 = 0$$

$$(x-9)(5x-2) = 0$$

Since x must be a whole number,

$$x = 9$$

So the total number of beads is

$$9 + 7 = 16$$

FINAL ANSWER

16 beads